

Irreducible rootless unimodular lattices generated by vectors of norm 3

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Abstract

The classical ADE classification implies that E_8 is the unique irreducible unimodular Euclidean lattice generated by vectors of norm 2. In sharp contrast, for every positive integer m , we construct an irreducible odd unimodular lattice of rank $24m$ and minimum norm 3 that is generated by its minimal vectors.

1 Introduction

Throughout, all lattices are Euclidean, and the norm of a lattice vector is its squared Euclidean length. Generation of lattices is always understood over $\mathbb{Z}$.

The classical ADE classification of root lattices states that every irreducible integral lattice generated by vectors of norm 2 is isometric to one of A_n ($n \geq 1$), D_n ($n \geq 4$), E_6 , E_7 , and E_8 ; see [6, 16]. Consequently, up to isometry, E_8 is the unique irreducible unimodular lattice generated by vectors of norm 2.

In this note, we show that the situation changes sharply at minimum norm 3. Even under the same requirements of irreducibility, unimodularity, and generation by minimal vectors, the rank is no longer bounded. More precisely, our main result is the following.

Theorem 1.1. *For every positive integer m , there exists an irreducible odd unimodular lattice L_m of rank $24m$ and minimum norm 3 that is 3-integrable and generated by its minimal vectors.*

Classical examples include the shorter Leech, odd Leech, and Conway–Borcherds lattices of ranks 23, 24, and 26, respectively [4, 8]. The irreducible unimodular lattices of minimum norm 3 in ranks 27 and 28 were classified by Bacher and Venkov [3], and those of rank 29 by Allombert and Chenevier [1]. All these lattices are generated by their minimal vectors; see [14] and [1, Example 4.3]. This property of generation is not automatic for unimodular lattices of minimum norm 3. For example, Chenevier [7, Proposition 8.2] gives a family of such lattices whose minimal vectors generate a proper finite-index sublattice in sufficiently large ranks.

Our construction cyclically glues copies of an index-3 sublattice of the odd Leech lattice. The constituent sublattices are generated by vectors of norm 3, and the added gluing vectors have the same norm. The key point is that this gluing introduces no vectors of smaller norm and produces

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an irreducible lattice for every $m \geq 1$. We establish these properties using codimension-one subcodes of extremal ternary self-dual codes of length 24.

As an application, we answer a question of Cao, Koolen, Liu, and Yang [5, Problem 1.9] affirmatively.

Corollary 1.2. *The lattices L_m yield pairwise switching-inequivalent connected non-extendable signed graphs Γ_m with smallest eigenvalue -3 .*

2 Preliminaries

A code C over $\mathbb{F}_3$ is called *ternary*. All codes in this paper are ternary linear codes. The *weight* $\text{wt}(\mathbf{c})$ of a word $\mathbf{c} \in \mathbb{F}_3^n$ is the number of its nonzero coordinates. A linear code of length n , dimension k , and minimum nonzero weight d is called an $[n, k, d]$ code. For a linear code C of length n , we write

$$A_i(C) := |\{\mathbf{c} \in C : \text{wt}(\mathbf{c}) = i\}| \quad (0 \leq i \leq n).$$

A code C of length n is said to be *self-dual* if $C = C^\perp$, where the dual code $C^\perp$ of C is defined as $C^\perp = \{\mathbf{x} \in \mathbb{F}_3^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \in C\}$. It was shown in [13] that the minimum weight d of a self-dual code of length n is bounded by $d \leq 3 \lfloor \frac{n}{12} \rfloor + 3$. If $d = 3 \lfloor \frac{n}{12} \rfloor + 3$, then the code is called *extremal*. Two codes C and C' are *equivalent* if there exists a monomial matrix P with $C' = C \cdot P = \{xP \mid x \in C\}$, and *inequivalent* otherwise.

Leon, Pless, and Sloane [12, Theorem 6] showed that there are precisely two equivalence classes of extremal ternary self-dual codes of length 24, represented by the extended ternary quadratic-residue code Q_{24} and the Pless symmetry code P_{24} .

The distinct supports of the weight-9 codewords of Q_{24} and P_{24} form two nonisomorphic 5-(24, 9, 6) designs [2, 15]. Consequently, by the above classification, the distinct supports of the weight-9 codewords of every extremal ternary self-dual $[24, 12, 9]$ code form a 5-(24, 9, 6) design. For $0 \leq j \leq 5$, let λ_j denote the number of blocks containing a fixed j -subset of the coordinate set. The derived parameters are

$$(\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (2024, 759, 264, 84, 24, 6). \quad (1)$$

Throughout, inner products of codewords are taken over $\mathbb{F}_3$, and we use $2 = -1$. If $\mathbf{c} \in \mathbb{F}_3^n$, its *canonical integral lift* $\tilde{\mathbf{c}} \in \{0, \pm 1\}^n$ is obtained by replacing the symbol 2 by -1 .

For later use, let

$$K_j(i) := \sum_{\ell=0}^j (-1)^\ell 2^{j-\ell} \binom{i}{\ell} \binom{24-i}{j-\ell} \quad (2)$$

be the ternary Krawtchouk polynomials of length 24. If E is a ternary $[24, r]$ code, $A_i = A_i(E)$, and $B_j = A_j(E^\perp)$, then

$$\sum_{i=0}^{24} A_i K_j(i) = 3^r B_j. \quad (3)$$

This is the Krawtchouk form of the MacWilliams identities; see [11, Chapter 7].

An n -dimensional lattice L is *integral* if $L \subseteq L^*$, where the dual lattice L^* is defined as $L^* = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} \in \mathbb{Z} \text{ for all } \mathbf{y} \in L\}$. A lattice L with $L = L^*$ is called *unimodular*. A unimodular lattice L is *even* if all vectors of L have even norms, and *odd* if some vector has an odd norm. A lattice L is called *s-integrable* if $\sqrt{s}L$ is isometric to a sublattice of $\mathbb{Z}^N$ for some positive integer N . A lattice is *irreducible* if it admits no orthogonal decomposition into two nonzero sublattices. A *root* is a vector of norm 2, and a lattice is *rootless* if it contains no roots. We write $\text{Min}(L)$ for the set of vectors attaining the minimum nonzero norm of L . Generation by minimal vectors means $L = \langle \text{Min}(L) \rangle_{\mathbb{Z}}$.

For a ternary linear code C of length n , its *Construction A lattice* is

$$\mathcal{A}_3(C) := \frac{1}{\sqrt{3}} \{ \mathbf{z} \in \mathbb{Z}^n : \mathbf{z} \bmod 3 \in C \}. \quad (4)$$

If C is self-dual and has minimum weight d , then $\mathcal{A}_3(C)$ is unimodular and 3-integrable with minimum norm $\min\{3, d/3\}$; see [10, Section 2]. In particular, $\mathcal{A}_3(P_{24})$ and $\mathcal{A}_3(Q_{24})$ are both isometric to the odd Leech lattice $\mathcal{O}_{24}$, the unique odd unimodular lattice of rank 24 and minimum norm 3; see [8, Chapter 17].

Lemma 2.1 ([10, Lemma 5]). *Let C be a ternary self-dual code of minimum weight at least 6. Then C is decomposable if and only if $\mathcal{A}_3(C)$ is decomposable.*

3 A ternary code seed

We start from an extremal ternary self-dual $[24, 12, 9]$ code C . Choose a word $\mathbf{w} \in C$ of weight 9. After a monomial transformation, we may assume $\mathbf{w} = \mathbf{1}_S$ and $|S| = 9$. Fix any partition $S = P \sqcup Q$ with $|P| = 4$ and $|Q| = 5$, and put

$$\mathbf{p} = -\mathbf{1}_P, \quad \mathbf{q} = -\mathbf{1}_Q.$$

Then

$$\mathbf{w} + \mathbf{p} + \mathbf{q} = \mathbf{0}, \quad \mathbf{p} \cdot \mathbf{p} = 1, \quad \mathbf{q} \cdot \mathbf{q} = 2, \quad \mathbf{p} \cdot \mathbf{q} = 0, \quad \mathbf{w} \cdot \mathbf{p} = 2.$$

Define

$$D = C \cap \mathbf{p}^\perp, \quad D_9 = \{ \mathbf{d} \in D : \text{wt}(\mathbf{d}) = 9 \}.$$

For a family of codewords, its *support graph* has the coordinate positions as vertices, with two coordinates adjacent when they occur together in the support of one member of the family.

Theorem 3.1. *For every choice of $C, \mathbf{w}, P, Q$ above, the following statements hold:*

- (i) $\dim D = 11$, $C = D \oplus \langle \mathbf{w} \rangle$, and $D \perp \mathbf{q}$;
- (ii) $D = \langle D_9 \rangle$;
- (iii) the support graph of D_9 is connected.

Proof. The functional $\mathbf{c} \mapsto \mathbf{c} \cdot \mathbf{p}$ is nonzero on C , since $\mathbf{w} \cdot \mathbf{p} = 2$. Hence its kernel D has dimension 11, and $C = D \oplus \langle \mathbf{w} \rangle$. Moreover, $D \perp \mathbf{p}$, and since $\mathbf{q} = -\mathbf{w} - \mathbf{p}$ and $D \subseteq C = C^\perp$, also $D \perp \mathbf{q}$. This proves (i).

Put $E = \langle D_9 \rangle$. We first show that E has full support. If $j \notin P$, inclusion-exclusion in the 5-(24, 9, 6) design shows that the number of blocks containing j and avoiding P is

$$\lambda_1 - 4\lambda_2 + 6\lambda_3 - 4\lambda_4 + \lambda_5 = 759 - 4 \cdot 264 + 6 \cdot 84 - 4 \cdot 24 + 6 = 117,$$

by (1). Every corresponding weight-9 word lies in D_9 . If $j \in P$, let ν_j be the number of blocks containing the 6-set $\{j\} \cup Q$. The number of blocks containing j and avoiding Q is

$$\lambda_1 - 5\lambda_2 + 10\lambda_3 - 10\lambda_4 + 5\lambda_5 - \nu_j = 69 - \nu_j \geq 63,$$

since $\nu_j \leq \lambda_5 = 6$. For a word $\mathbf{c}$ on such a block, $\mathbf{c} \cdot \mathbf{w} = 0$ and $\mathbf{c}$ vanishes on Q , whence $\mathbf{c} \cdot \mathbf{p} = 0$. Thus every coordinate occurs in a word of D_9 .

We next establish a lower bound on $|D_9|$. Let $a = A_9(D)$ and $B_j = A_j(D^\perp)$. Since $D^\perp = C^\perp + \langle \mathbf{p} \rangle = C + \langle \mathbf{p} \rangle$, the only nonzero words of $D^\perp$ of weight at most 4 are $\mathbf{p}$ and $-\mathbf{p}$. Indeed, if $\mathbf{c} \in C$ and $\alpha \neq 0$ satisfy $\text{wt}(\mathbf{c} + \alpha \mathbf{p}) \leq 4$, then $\text{wt}(\mathbf{c}) \leq 8$, so $\mathbf{c} = \mathbf{0}$. Hence

$$B_0 = 1, \quad B_1 = B_2 = B_3 = 0, \quad B_4 = 2.$$

Since $D \subset C$, all weights in D are divisible by 3, and its nonzero weights are at least 9. Substitution in the standard identities (3) for $j = 0, 1, 2, 3, 4$ gives

$$\begin{aligned} A_{12}(D) &= 27362 - 5a, \quad A_{15}(D) = 67400 + 10a, \quad A_{18}(D) = 79748 - 10a, \\ A_{21}(D) &= 1240 + 5a, \quad A_{24}(D) = 1396 - a. \end{aligned}$$

The equation for $j = 5$ then gives $B_5 = \frac{a-1360}{3}$. Consequently

$$|D_9| = a \geq 1360. \quad (5)$$

It remains to show that $\dim E = 11$. We use the following Delsarte-type linear-programming argument [9]. Set

$$F(i) = \frac{1532 + 404K_1(i) - 53K_5(i) - 11K_6(i)}{228906}. \quad (6)$$

Direct evaluation from (2) gives

i	0	9	12	15	18	21	24
$F(i)$	$-\frac{114418}{157}$	1	0	0	0	$\frac{1}{314}$	$\frac{524}{157}$

(7)

Let $r = \dim E$ and write $B'_j = A_j(E^\perp)$. Since E has full support, $B'_1 = 0$. Since the nonzero weights of E belong to $\{9, 12, 15, 18, 21, 24\}$, (7) gives

$$\sum_{i>0} A_i(E)F(i) = A_9(E) + \frac{1}{314}A_{21}(E) + \frac{524}{157}A_{24}(E) \geq A_9(E).$$

On the other hand, by (6), the MacWilliams identities (3) yield

$$\sum_{i=0}^{24} A_i(E)F(i) = \frac{3^r}{228906} (1532B'_0 + 404B'_1 - 53B'_5 - 11B'_6) \leq \frac{1532 \cdot 3^r}{228906},$$

because $B'_0 = 1$, $B'_1 = 0$, and $B'_5, B'_6 \geq 0$. Since $A_0(E) = 1$ and $F(0) = -114418/157$, it follows that

$$A_9(E) \leq \sum_{i>0} A_i(E)F(i) \leq \frac{1532 \cdot 3^r}{228906} + \frac{114418}{157}.$$

Note that $A_9(E) = |D_9| \geq 1360$ by (5), and when $r \leq 10$, $A_9(E) < 1360$. Thus $r \geq 11$. Since $E \subseteq D$ and $\dim D = 11$, we have $E = D$. This proves (ii).

For (iii), if the support graph had $t \geq 2$ components, generation would give $D = \bigoplus_{i=1}^t D_i$, with nonzero $[n_i, k_i, d_i]_3$ codes satisfying $d_i \geq 9$. The Singleton bound gives

$$13 = 24 - 11 = \sum_i (n_i - k_i) \geq \sum_i (d_i - 1) \geq 8t \geq 16,$$

a contradiction. This completes the proof. □

Lemma 3.2. For $a, b \in \mathbb{F}_3$, put $\mu(a, b) = \min\{\text{wt}(\mathbf{d} + a\mathbf{q} + b\mathbf{p}) : \mathbf{d} \in D\}$. With rows and columns indexed by $0, 1, 2$, the following inequality holds entrywise:

$$(\mu(a, b))_{a, b \in \mathbb{F}_3} \geq \begin{pmatrix} 0 & 4 & 4 \\ 5 & 9 & 6 \\ 5 & 6 & 9 \end{pmatrix} \quad (8)$$

Proof. Let $\mathbf{u} = \mathbf{d} + a\mathbf{q} + b\mathbf{p}$ with $\mathbf{d} \in D$. Since D is orthogonal to both $\mathbf{p}$ and $\mathbf{q}$, and D is self-orthogonal,

$$\text{wt}(\mathbf{u}) \equiv \mathbf{u} \cdot \mathbf{u} \equiv 2a^2 + b^2 \pmod{3}. \quad (9)$$

The first row and column follow from $\text{wt}(\mathbf{p}) = 4$, $\text{wt}(\mathbf{q}) = 5$, $d(C) = 9$, and (9). If $a = b \neq 0$, then $a\mathbf{q} + b\mathbf{p} = -a\mathbf{w}$, so $\mathbf{d} - a\mathbf{w}$ is a nonzero word of C and has weight at least 9. If $a = -b \neq 0$, scalar multiplication reduces to $(a, b) = (1, 2)$. For $\mathbf{u} = \mathbf{d} + \mathbf{q} - \mathbf{p}$, the word $\mathbf{u} - \mathbf{p} = \mathbf{d} - \mathbf{w}$ lies nontrivially in C , so $\text{wt}(\mathbf{u}) \geq 9 - 4 = 5$; by (9) its weight is divisible by 3, and hence is at least 6. □

4 Construction of the lattices L_m

For every integer $m \geq 1$, we first construct an indecomposable ternary self-dual $[24m, 12m, 9]$ code C_m generated by its minimum-weight codewords.

Fix $C, \mathbf{w}, P, Q, \mathbf{p}, \mathbf{q}, D$ as in section 3. For $m \geq 1$, index m blocks of length 24 by $i \in \mathbb{Z}/m\mathbb{Z}$. For $\mathbf{v} \in \mathbb{F}_3^{24}$, write $\mathbf{v}^{(i)}$ for the vector supported on block i with value $\mathbf{v}$. Define

$$\mathbf{x}_i = \mathbf{p}^{(i)} + \mathbf{q}^{(i+1)}, \quad X_m = \langle \mathbf{x}_i : i \in \mathbb{Z}/m\mathbb{Z} \rangle, \quad C_m = D^{\oplus m} + X_m. \quad (10)$$

For $m = 1$, $\mathbf{x}_0 = \mathbf{p} + \mathbf{q} = -\mathbf{w}$, so $C_1 = C$.

Theorem 4.1. *For every $m \geq 1$, C_m is an indecomposable ternary self-dual code with parameters $[24m, 12m, 9]$. It is generated by weight-9 words whose support graph is connected.*

Proof. The code $D^{\oplus m}$ is self-orthogonal and is orthogonal to every bridge. Moreover, $\mathbf{x}_i \cdot \mathbf{x}_i = 0$, and distinct bridges are orthogonal. Hence C_m is self-orthogonal.

If $\sum_i \alpha_i \mathbf{x}_i \in D^{\oplus m}$, then its block- i component is $\alpha_i \mathbf{p} + \alpha_{i-1} \mathbf{q}$. Taking its inner product with $\mathbf{p}$ gives $\alpha_i = 0$. Thus the m bridges are independent and $X_m \cap D^{\oplus m} = 0$. Therefore $\dim C_m = 11m + m = 12m$, so C_m is self-dual.

Write a word of C_m as $\mathbf{z} = (\mathbf{d}_i)_{i \in \mathbb{Z}/m\mathbb{Z}} + \sum_i \alpha_i \mathbf{x}_i$. Its block- i component is $\mathbf{z}_i = \mathbf{d}_i + \alpha_{i-1} \mathbf{q} + \alpha_i \mathbf{p}$. The weight of $\mathbf{z}$ satisfies $\text{wt}(\mathbf{z}) = \sum_i \text{wt}(\mathbf{z}_i) \geq \sum_i \mu(\alpha_{i-1}, \alpha_i)$. Therefore, the matrix (8) assigns a lower-bound cost to each transition $\alpha_{i-1} \rightarrow \alpha_i$ in the cyclic state sequence. If all states are zero, every nonzero word has weight at least 9. If both zero and nonzero states occur, every maximal nonzero run has entrance cost at least 4 and exit cost at least 5. If all states are nonzero, either they are all equal, in which case a transition already costs at least 9, or both 1 and 2 occur, in which case the cycle has at least two transitions of cost at least 6. Hence $d(C_m) \geq 9$. Each bridge has weight 9, so $d(C_m) = 9$.

By Theorem 3.1, the block-local copies of D_9 , together with the bridges, generate C_m ; all have weight 9. Within each block their support graph is connected, and every bridge joins two consecutive blocks. Thus the full support graph is connected.

Finally, suppose that C_m decomposes over a nontrivial coordinate partition as $U \oplus V$. Since $d(C_m) = 9$, a weight-9 word cannot have nonzero projections in both summands. This contradicts the connectedness of the support graph. Thus C_m is indecomposable. $\square$

We now prove Theorem 1.1 by taking $L_m := \mathcal{A}_3(C_m)$ to be the Construction A lattice associated with C_m . By (4), L_m is 3-integrable.

Proof of Theorem 1.1. The Construction A properties recalled in Section 2 show that L_m is unimodular of rank $24m$ and minimum norm 3. Lemma 2.1 gives irreducibility, and a vector of norm 3 shows that L_m is odd.

The vectors $\sqrt{3}\mathbf{e}_j$ are minimal and generate $\sqrt{3}\mathbb{Z}^{24m}$. For every weight-9 word $\mathbf{c} \in C_m$, its canonical lift $\tilde{\mathbf{c}}/\sqrt{3}$ is also minimal. Since the weight-9 words generate C_m and $L_m/\sqrt{3}\mathbb{Z}^{24m} \cong C_m$, these minimal vectors generate L_m . This completes the proof. $\square$

Remark 4.2. *A computer-assisted classification shows that, for each fixed $m \geq 2$, our construction yields exactly four isometry classes of lattices, listed in Table 1. Put $a := A_9(D)$ and $b := |\{\mathbf{c} \in D + \mathbf{q} : \text{wt}(\mathbf{c}) = 5\}|$. Their kissing numbers are given by*

$$|\text{Min}(L_m)| = 48m + A_9(C_m) = m(48 + a + 2b),$$

and distinguish the four classes. For $m = 1$, all choices give the odd Leech lattice.

Table 1: The four lattice isometry classes arising from the extremal $4 + 5$ construction for each fixed $m \geq 2$.

Type	$a = A_9(D)$	b	$ \text{Min}(L_m) $	Starting code C
I	1372	2	$1424m$	Q_{24} or P_{24}
II	1378	3	$1432m$	Q_{24} or P_{24}
III	1384	4	$1440m$	Q_{24} or P_{24}
IV	1396	6	$1456m$	P_{24}

All lattices constructed here are 3-integrable. This suggests the following question.

Question 4.3. *Does there exist a positive integer t such that every integral lattice generated by vectors of norm 3 is t -integrable?*

5 Non-extendable signed graphs

A *signed graph* is a finite simple graph G together with a map $\sigma : E(G) \rightarrow \{\pm 1\}$. Its signed adjacency matrix $A = A(G, \sigma)$ has entry $\sigma(\{x, y\})$ when x and y are adjacent, and 0 otherwise. Its eigenvalues are those of A . Switching replaces A by TAT , where T is diagonal with entries in $\{\pm 1\}$. We consider switching classes up to vertex relabelling. Following [5, Definition 1.4], a connected signed graph with smallest eigenvalue -3 is *non-extendable* if it is not a proper induced signed subgraph of any connected signed graph with smallest eigenvalue at least -3 .

We prove Corollary 1.2 by constructing Γ_m from one representative of each antipodal pair of minimal vectors of L_m . To prove non-extendability, we use a projection argument similar to that in the proof of [5, Lemma 2.2].

Proof of Corollary 1.2. Choose one representative from each antipodal pair $\{\mathbf{v}, -\mathbf{v}\} \subseteq \text{Min}(L_m)$, and denote the resulting set by S_m . For distinct $\mathbf{u}, \mathbf{v} \in S_m$, the vectors $\mathbf{u} \pm \mathbf{v}$ are nonzero, so $3 \leq (\mathbf{u} \pm \mathbf{v}) \cdot (\mathbf{u} \pm \mathbf{v}) = 6 \pm 2\mathbf{u} \cdot \mathbf{v}$. Integrality therefore gives $\mathbf{u} \cdot \mathbf{v} \in \{0, \pm 1\}$. Define a signed graph Γ_m on S_m by joining distinct vectors when their inner product is nonzero, with this inner product as the edge sign. Writing $A_m = A(\Gamma_m)$, we have $3I + A_m = \text{Gram}(S_m) \succeq 0$. Since S_m generates L_m , $\text{rank}(3I + A_m) = 24m$. There are $24m$ antipodal pairs of coordinate minimal vectors $\pm\sqrt{3}\mathbf{e}_j$, and at least one further pair arising from a weight-9 word of C_m . Hence $|S_m| > 24m$, so $3I + A_m$ is singular and $\lambda_{\min}(\Gamma_m) = -3$. Different choices of representatives only switch Γ_m . Moreover, if Γ_m were disconnected, the integer spans of its components would give a nontrivial orthogonal decomposition of L_m . Thus Γ_m is connected.

Suppose that a connected signed graph Γ' properly contains Γ_m as an induced signed subgraph and has smallest eigenvalue at least -3 . Choose a Gram representation of $3I + A(\Gamma')$. Identify the span of the old Gram vectors isometrically with $V = \text{span}_{\mathbb{R}} L_m$, so that the old vectors are precisely those of S_m . Since Γ' is connected, some new vertex is adjacent to an old vertex. Let $\mathbf{y}$ be its Gram vector and let $\mathbf{z}$ be its orthogonal projection onto V . Then $\mathbf{z} \neq 0$ and $\mathbf{z} \cdot \mathbf{v} = \mathbf{y} \cdot \mathbf{v} \in \{0, \pm 1\}$ for every $\mathbf{v} \in S_m$. As S_m generates L_m , we have $\mathbf{z} \in L_m^* = L_m$. The minimum norm and the projection inequality give

$$3 \leq \mathbf{z} \cdot \mathbf{z} \leq \mathbf{y} \cdot \mathbf{y} = 3.$$

Equality forces $\mathbf{y} = \mathbf{z} \in \text{Min}(L_m)$, and hence $\mathbf{y} = \pm \mathbf{v}$ for some $\mathbf{v} \in S_m$. This gives the impossible off-diagonal Gram entry $\mathbf{y} \cdot \mathbf{v} = \pm 3$. Therefore Γ_m is non-extendable.

Finally, $\text{rank}(A_m + 3I) = 24m$ is invariant under switching and vertex relabelling. Different values of m therefore give pairwise distinct switching classes. $\square$

Statement of AI use. ChatGPT 5.6 Sol initially provided a specific ternary self-dual [24, 12, 9] code, together with a computer-assisted argument showing that this code gives rise to an infinite

family of lattices. Building on this, we generalized the construction to arbitrary ternary self-dual $[24, 12, 9]$ codes and gave a proof that does not rely on computer-assisted verification. The authors take full responsibility for the mathematical claims, proofs, and final presentation of the paper.

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